# DIGITAL PAYMENT USAGE PATTERNS AND FINANCIAL BEHAVIOR: A STATISTICAL ANALYSIS OF CONSUMER DEMOGRAPHICS

PROJECT REPORT

The project report submitted for the partial fulfilment of the requirement

award of the degree of

**BACHELOR OF SCIENCE IN STATISTICS**

**UNIVERSITY OF CALICUT**

SUBMITTED BY

……..

**UNDER THE SUPERVISION AND GUIDANCE OF**

**Ms.AMARSWATHA**

**CHMKM GOVT ARTS AND SCIENCE COLLEGE KODUVALLY**

(AFFILIATED TO THE CALICUT UNIVERSITY)

2022-2025

**DEPARTMENT OF STATISTICS**

## CHMKM GOVT.ARTS AND SCIENCE COLLEGE KODUVALLY

# CERTIFICATE

This is to certify that ---------- is a bonafide student of the Department of Statistics, CHMKM Government Arts and Science College, Koduvally, and that this project report entitled "**Digital Payment Usage Patterns and Financial Behavior: A Statistical Analysis of Consumer Demographics**" has been prepared by in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Statistics under the University of Calicut.

**SWAPNA A J** **AMARSWATHA**

(HOD of Statistics & Computer Science) (Project Guide)

**External Examiner:**

# DECLARATION

I, ……, hereby declare that this project entitled **"Digital Payment Usage Patterns and Financial Behavior: A Statistical Analysis of Consumer Demographics"** is my original work, completed under the guidance of **Ms. Amarswatha, Department of Statistics, CHMKM Government Arts and Science College, Koduvally.**

The results embodied in this project have not been included any other dissertation Submitted previously for the award of any degree or diploma of any other university or institution.

Place: Name:

Date: Signature:

# DIGITAL PAYMENT USAGE PATTERNS AND FINANCIAL BEHAVIOR: A STATISTICAL ANALYSIS OF CONSUMER DEMOGRAPHICS

Project submitted in partial fulfilment of requirement to the award of degree

B.Sc. Statistics.

PREPARED BY

**SREYA P(KXAWSST014)**

**THAHIRA HUSNA P A(KXAWSST015)**

**SOURAV P K(KXAWSST019)**

**AKHIL(KXAWSST024)**

GUIDED BY

**Ms. AMARSWATHA**

# ACKNOWLEDGEMENT

At the successful completion of our project, we take this opportunity to express our sincere gratitude to everyone who supported and guided us throughout this journey.

First and foremost, we are thankful to **Almighty God** for showering us with His blessings, strength, and guidance to complete this work successfully.

We extend our heartfelt thanks to **Asst. Prof. Suresh K.M., Principal, CHMKM Govt. Arts and Science College, Koduvally**, for his constant encouragement and support.

We express our deep gratitude to **Ms.Swapna A J, Head of the Department of Statistics**, for her valuable suggestions and guidance during the development of this project.

Our special thanks go to **Ms.Amarswatha, our project guide**, for her dedicated guidance, timely advice, and unwavering support that helped us successfully complete this work.

Finally, we express our sincere appreciation to all the **faculty members of the Department of Statistics** for their continuous guidance, encouragement, and help at every stage of this project.

Name:

Signature:

# CONTENTS

**CHAPTER 1**

**INTRODUCTION……7**

1.1 Background of the Study……..7

1.2 Objectives of the Study…...……..…..…8

**CHAPTER 2**

**METHODOLOGY……9**

2.1 Graphical Tools……...……9

2.2 Software Used……...……9

2.3 Statistical Tools……10

**CHAPTER 3**

**DATA ANALYSIS……14**

3.1 Graphical Analysis……..…14

3.2 Statistical Analysis……..…31

**CHAPTER 4**

**CONCLUSION……44**

**BIBLIOGRAPHY……45**

**QUESTIONNAIRE……46**

# CHAPTER 1

# INTRODUCTION

## 1.1 Background of the Study

The rise of digital payment systems has revolutionized the way consumers conduct financial transactions, significantly altering financial behavior and payment preferences. With the increasing adoption of smartphones and the availability of high-speed internet, digital payment platforms have become integral to everyday life. From paying utility bills to shopping and entertainment, these systems offer convenience, speed, and enhanced security, making them a preferred choice for millions of users worldwide.

This study explores the patterns of digital payment usage, and the financial behaviors associated with them, focusing on consumer demographics. The objective is to understand how factors such as age, gender, educational qualification, occupation, and income influence the adoption and frequency of digital payment methods. Additionally, the study seeks to uncover user preferences for specific platforms, the types of transactions commonly conducted, and perceptions of benefits such as expense tracking, savings, and convenience.

By analyzing user satisfaction, trust levels, and challenges such as transaction failures or security concerns, this research aims to provide a comprehensive overview of the evolving landscape of digital payments. Furthermore, the study examines the broader implications of digital payments on financial management, including impulse purchases, timely bill payments, and overall consumer confidence in the digital ecosystem.

The findings from this analysis will contribute to a deeper understanding of consumer behavior in the digital payment domain, offering valuable insights for policymakers, financial institutions, and digital payment providers to enhance user experience and promote financial inclusion.

## 1.2 Objectives of the Study

- To analyse the association between demographic characteristics and digital payment adoption patterns.
- To evaluate the impact of digital payments on personal financial management.
- To assess user experience factors influencing digital payment usage.
- To examine security perceptions and their influence on digital payment usage patterns.
- To analyse platform preferences for different transaction types and frequency of use.

# CHAPTER 2

# METHODOLOGY

## 2.1 Graphical Tools

### *Bar Chart*

A bar chart is a simple graphical tool that uses rectangular bars to represent data. Each bar reflects the value of a category, making it easy to compare data across groups. Categories are shown on the horizontal axis, and values on the vertical axis. Bar charts are ideal for showing comparisons, distributions, and trends in discrete data, offering a clear and easy way to interpret patterns briefly.

### *Pie Chart*

A pie chart is a circular graph divided into slices to show each category's proportion within a whole. Each slice represents a percentage of the total, making it easy to visualize parts of a whole. Pie charts are ideal for showing percentages, ratios, and compositions, but are less effective when there are too many categories or when precise comparisons are needed.

## 2.2 Software Used

### *Excel*

Excel is used for spread sheet creation and to do simple data analysis. Spreadsheet is a collection of cells where each cell is intersection of row and column identified by its column and row index. For instance, the cell belonging to Column A and Row I is denoted by Al. Cell accepts descriptive Text/Numbers/Formula. The cell which you are presently working at is called active or current cell.

### *MS Word*

Microsoft Word is a word-processing software used to create, edit, format, and print text documents. It allows users to write letters, reports, resumes, assignments, and more. Word offers various tools like text formatting, tables, images, page layout, spell check, and styles to make documents professional and organized.

### *SPSS*

It is powerful software used for data analysis, statistical tests, and data management. It helps researchers handle large datasets, perform statistical analysis, and create graphs easily. SPSS is widely used in social sciences, healthcare, marketing, education, and business research to analyze data and draw meaningful conclusions.

### *Google Forms*

Google Form is a free online tool provided by Google that allows users to create surveys, quizzes, and forms for data collection. It enables easy sharing, automatic response collection, and real-time data organization in Google Sheets, making it an effective method for gathering and analysing information efficiently.

## 2.3 Statistical Tools

### *Mean*

Mean is a measure of central tendency that represents the average value of a given set of data. It is calculated by dividing the sum of all observations by the total number of observations. The mean provides a single value that summarizes the entire dataset and is widely used for comparing and analysing data.

Formula:

$$\text{Mean } (\overline{x}) = \frac{\sum_{i=1}^{n} x_i}{n}$$

## *Chi-Square Test of Independence*

The Chi-square test of independence is a statistical test used to determine if there is a significant association between two categorical variables. It compares the observed and expected frequencies in a contingency table. If the difference between them is large, it suggests that the variables are related. The test provides a p-value, and if this p-value is less than the chosen significance level, we conclude that there is a significant association. Chi-square is widely used to analyze relationships between categorical data and draw inferences about their dependence.

Formula:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

Where:

Oi -Observed Frequency

Ei -Expected frequency

## *Correlation*

Correlation is a statistical measure that shows the strength and direction of a relationship between two variables. Its value ranges from -1 to 1, where -1 indicates a perfect negative relationship, 1 a perfect positive relationship, and 0 means no linear relationship. A positive correlation means both variables move in the same direction, while a negative correlation means they move in opposite directions. The correlation coefficient (r) indicates how strong or weak this relationship is.

Formula:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where:

r = Correlation coefficient

xi = Value of the x-variable in a sample

$\bar{x}$= Mean of the x-variable

yi = Value of the y-variable in a sample

$\bar{y}$ = Mean of the y-variable

## *The Mann-Whitney U Test*

The Mann-Whitney U test is a non-parametric test used to compare two independent groups when data are not normally distributed. It tests whether there is a significant difference in the distributions or medians of the groups. It serves as an alternative to the independent samples t-test when normality assumptions are violated.

Formula:

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

Where:

$n_1$, $n_2$= Sizes of group 1 and group 2.

$R_1$, $R_2$ = Sum of ranks for group 1 and group 2.

$U_1$, $U_2$=U statistics for each group.

## *Kruskal-Wallis Test*

The Kruskal-Wallis test is a non-parametric alternative to ANOVA, used to determine if there are significant differences between the medians of three or more independent groups when data do not meet normality assumptions. It extends the Mann-Whitney U test to multiple groups by ranking all data points together and comparing the sum of ranks across groups. This test is suitable when the dependent variable is ordinal or continuous, and the independent variable has three or more groups.

Formula:

$$H = \left[ \frac{12}{n(n+1)} \sum_{j=1}^{c} \frac{T_j^2}{n_j} \right] - 3(n+1)$$

Where:

$n$ = Total number of observations

$c$ = Number of groups

$T_j$= Sum of ranks for group j

$n_j$= Number of observations in group j

## *P-Value*

P-value is a probability value used in hypothesis testing to determine the significance of results. It indicates the likelihood of obtaining the observed data (or more extreme) assuming the null hypothesis is true. A small p-value (typically $\leq 0.05$) suggests strong evidence against the null hypothesis, leading to its rejection. A large p-value ($> 0.05$) indicates weak evidence against the null hypothesis, so we fail to reject it.

## *Questionnaire*

It is a research instrument consisting of a series of structured questions designed to collect information, opinions, or attitudes from respondents. It is widely used in surveys to gather both quantitative and qualitative data on various topics. A questionnaire may include open-ended or close-ended questions and helps in obtaining standardized and comparable responses from a large number of participants efficiently.

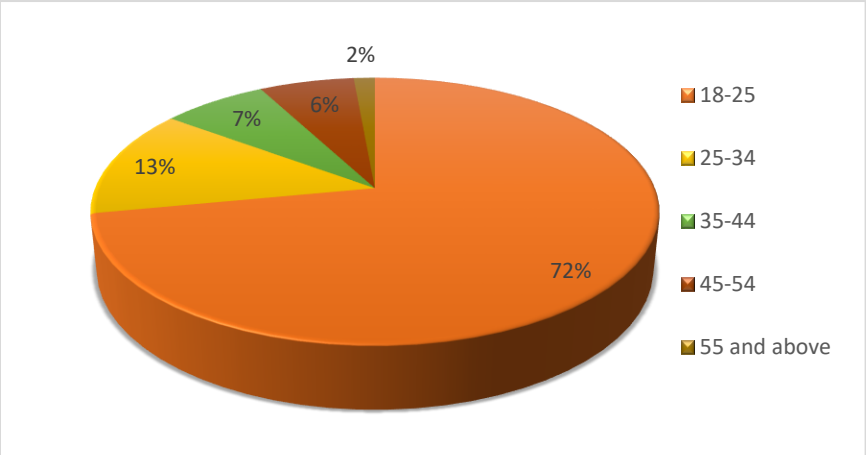
# CHAPTER 3

# DATA ANALYSIS

## 3.1 Graphical Analysis

### *3.1.1 Age-wise Distribution*

| Row Labels | Count of Age Group |
|---|---|
| 18-25 | 203 |
| 25-34 | 37 |
| 35-44 | 20 |
| 45-54 | 18 |
| 55 and above | 4 |
| Grand Total | 282 |

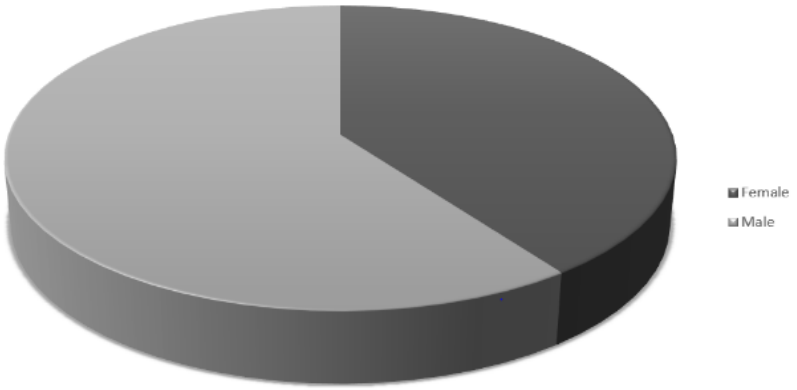



***Inferences:***

The pie chart shows that the majority (72%) of the respondents are aged 18-25, indicating high interest or engagement from this age group. Interest significantly decreases with age, with the 25-34 group at 13%, 35-44 at 7%, 45-54 at 6%, and 55 and above at 2%. The data suggests that the younger demographic is the primary audience.

### *3.1.2 Gender-wise Distribution*

| Row Labels | Count of Gender |
|---|---|
| **Female** | 113 |
| **Male** | 169 |
| **Grand Total** | **282** |

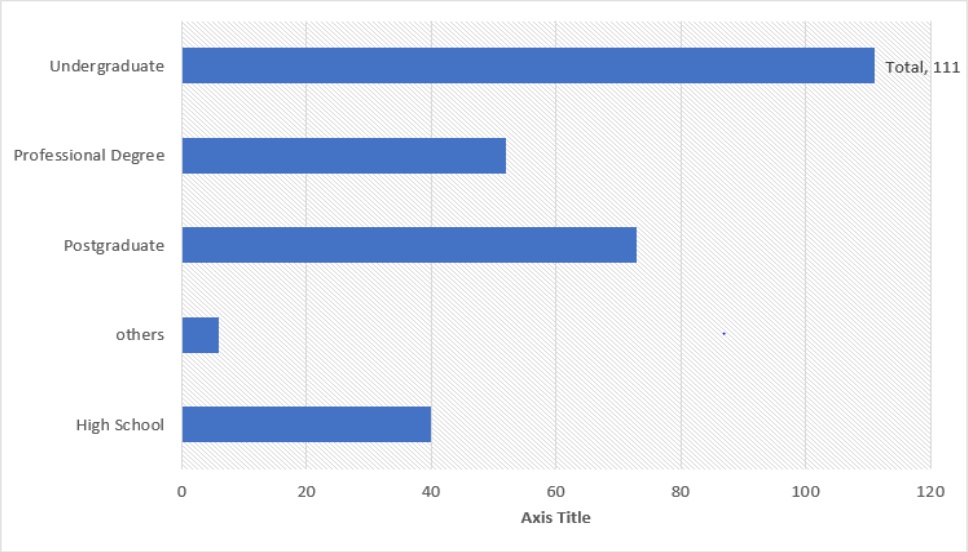



***Inferences:***

The image shows a gender distribution report with a table and a pie chart. The table indicates that out of a total of 282 individuals, 113 are female and 169 are male. The pie chart visually represents this distribution, showing that males form a larger portion than females.

### *3.1.3 Education-wise Distribution*

| Row Labels | Count of Education |
|---|---|
| **High School** | 40 |
| **others** | 6 |
| **Postgraduate** | 73 |
| **Professional Degree** | 52 |
| **Undergraduate** | 111 |
| **Grand Total** | **282** |

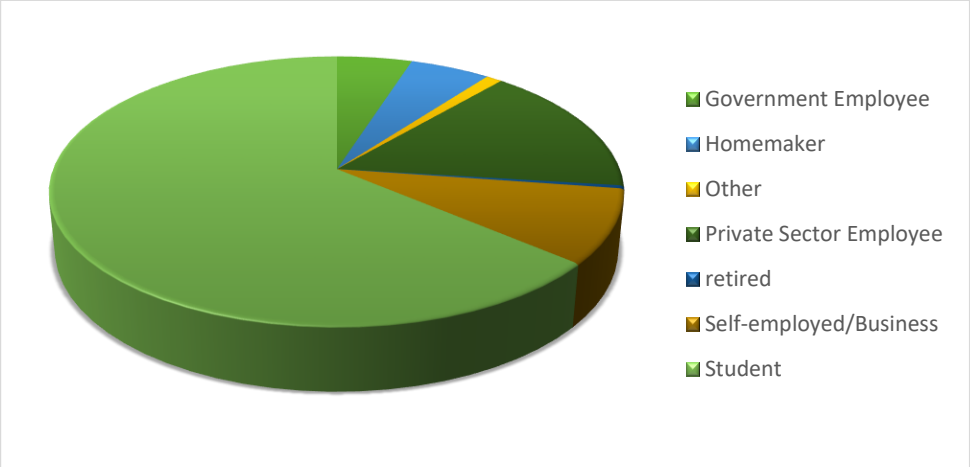



### *Inferences:*

The data on education shows that the majority of individuals have an undergraduate degree (111), followed by postgraduate degrees (73), professional degrees (52), and high school education (40). A small group (6) falls into the "others" category, totaling 282 respondents.

### *3.1.4 Occupation-wise Distribution*

| Row Labels | Count of Occupation |
|---|---|
| Government Employee | 14 |
| Homemaker | 15 |
| Other | 3 |
| Private Sector Employee | 44 |
| retired | 1 |
| Self-employed/Business | 25 |
| Student | 180 |
| Grand Total | 282 |

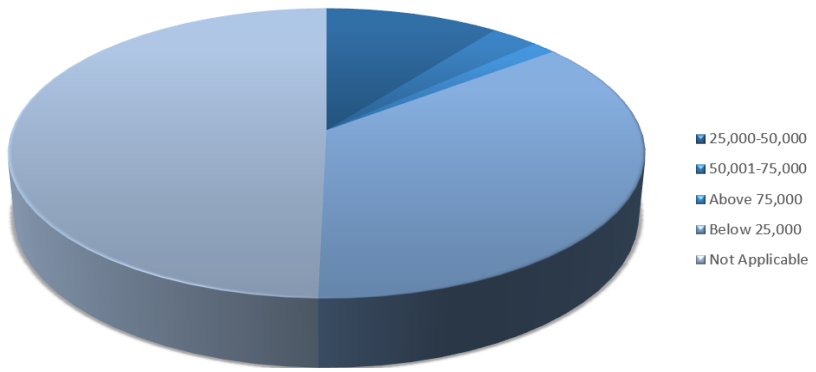



***Inferences:***

The data shows the distribution of occupations among 282 respondents. The largest group is students (180), followed by private sector employees (44) and self-employed/business individuals (25). Homemakers (15) and government employees (14) are also notable. Other categories like retired, unemployed, and temporary government employees have very low counts (1 each).

### *3.1.5 Monthly Income-wise Distribution*

| Row Labels | Count of Monthly Income |
|---|---|
| 25,000-50,000 | 29 |
| 50,001-75,000 | 8 |
| Above 75,000 | 4 |
| Below 25,000 | 101 |
| Not Applicable | 140 |
| Grand Total | 282 |

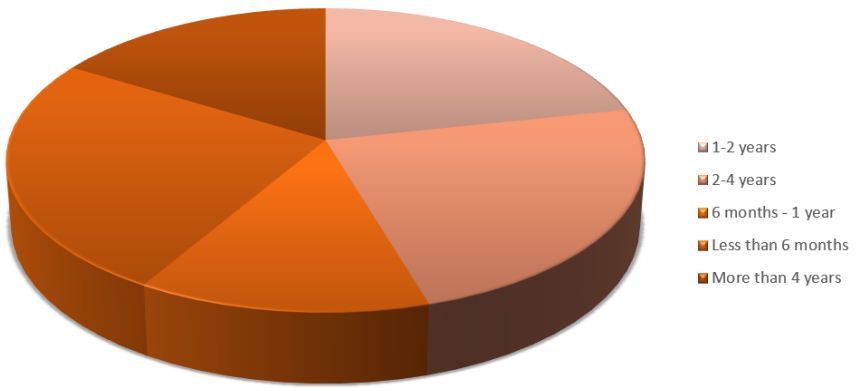



***Inferences:***

The largest group is "Not Applicable" (140 respondents, 49.6%), likely including students or unemployed individuals. The second largest is "Below 25,000" (101 respondents, 35.8%), indicating a significant low-income segment. Moderate income (25,000 - 50,000) includes 29 respondents (10.3%). Higher income groups, 50,001 - 75,000 (8 respondents, 2.8%) and Above 75,000 (4 respondents, 1.4%), much are smaller.

### *3.1.6 Digital Payment Usage Experience*

| Row Labels | Count of Digital Payment Experience |
|---|---|
| 1-2 years | 61 |
| 2-4 years | 67 |
| 6 months - 1 year | 37 |
| Less than 6 months | 71 |
| More than 4 years | 46 |
| Grand Total | **282** |

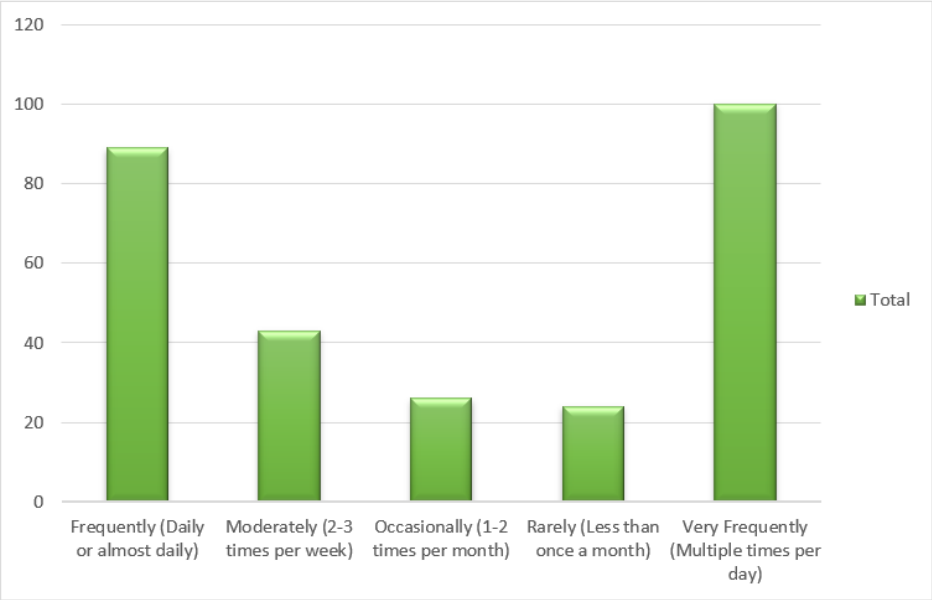



***Inferences:***

The table and pie chart show digital payment experience across five categories. The most common experience range is "Less than 6 months" with 71 counts, while the least common is "More than 4 years" with 46 counts. The total responses are 282, with most respondents having less than 2 years of experience.

### *3.1.7 Frequency of Using Digital Payment*

| Row Labels | Count of Usage Frequency |
|---|---|
| Frequently (Daily or almost daily) | 89 |
| Moderately (2-3 times per week) | 43 |
| Occasionally (1-2 times per month) | 26 |
| Rarely (Less than once a month) | 24 |
| Very Frequently (Multiple times per day) | 100 |
| **Grand Total** | **282** |

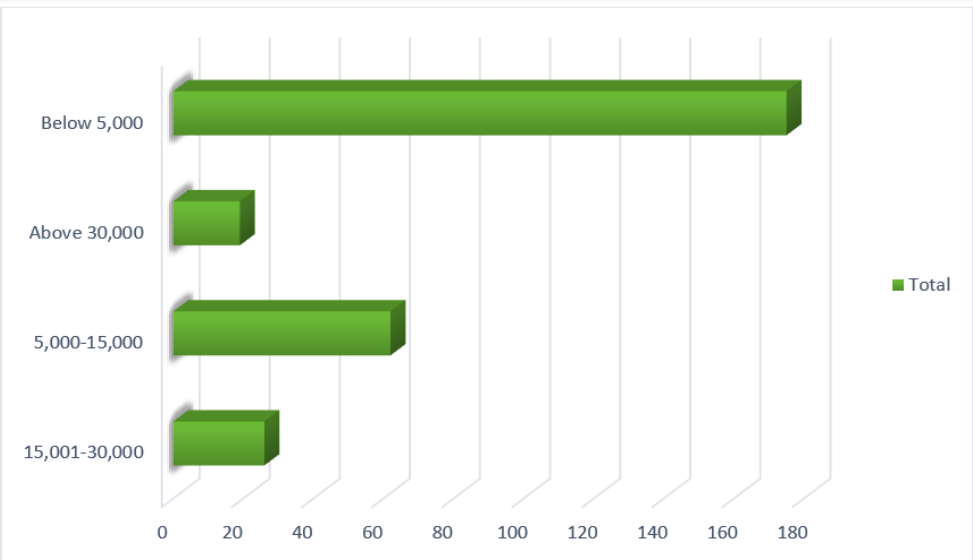



***Inferences:***

Digital payment usage is highest for "Very Frequently" (100 counts) and lowest for "Rarely" (24 counts). Usage generally declines as frequency decreases, except for a sharp rise in the highest category.

### *3.1.8 Average Monthly Transaction Count*

| Row Labels | Count of Avg Monthly Transactions |
|---|---|
| 15,001-30,000 | 26 |
| 5,000-15,000 | 62 |
| Above 30,000 | 19 |
| Below 5,000 | 175 |
| Grand Total | 282 |

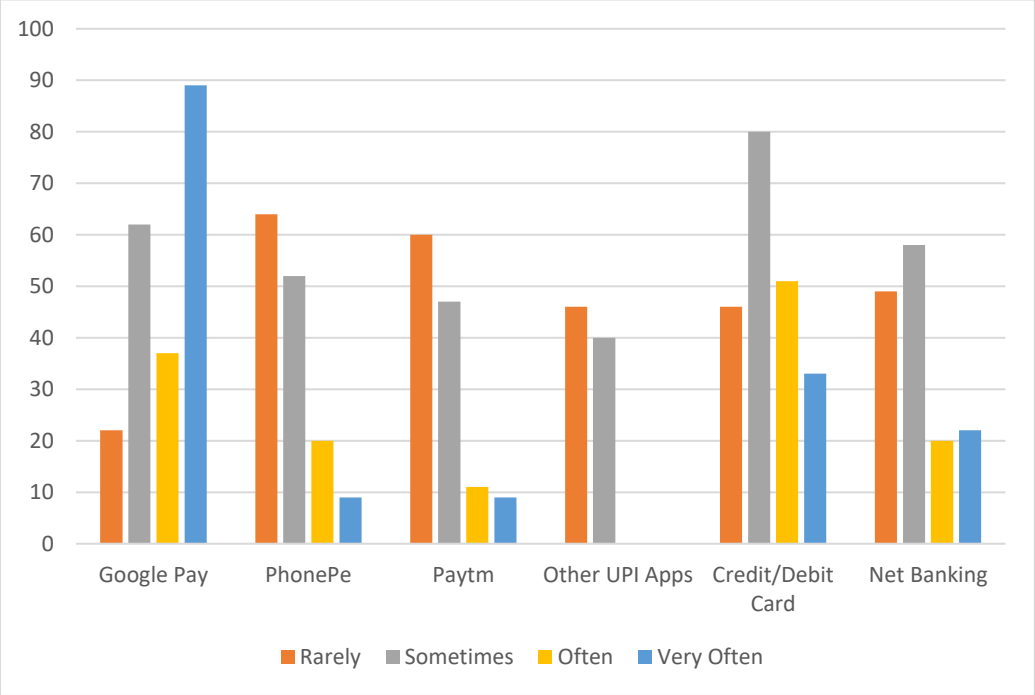



***Inferences:***

The table and chart show the count of average monthly transactions across five categories. The majority of respondents (175 out of 282) have transactions below 5,000, while the fewest (19 respondents) have transactions above 30,000. Most transactions fall within lower ranges, indicating limited or smaller-scale usage.

### *3.1.9 Frequency of Usage Across Different Digital Payment Methods*

| Usage | Google Pay | PhonePe | Paytm | Other UPI Apps | Credit/Debit Card | Net Banking |
|---|---|---|---|---|---|---|
| **Never** | 34 | 116 | 121 | 120 | 72 | 133 |
| **Rarely** | 22 | 64 | 60 | 46 | 46 | 49 |
| **Sometimes** | 62 | 52 | 47 | 40 | 80 | 58 |
| **Often** | 37 | 20 | 11 | 0 | 51 | 20 |
| **Very Often** | 89 | 9 | 9 | 0 | 33 | 22 |

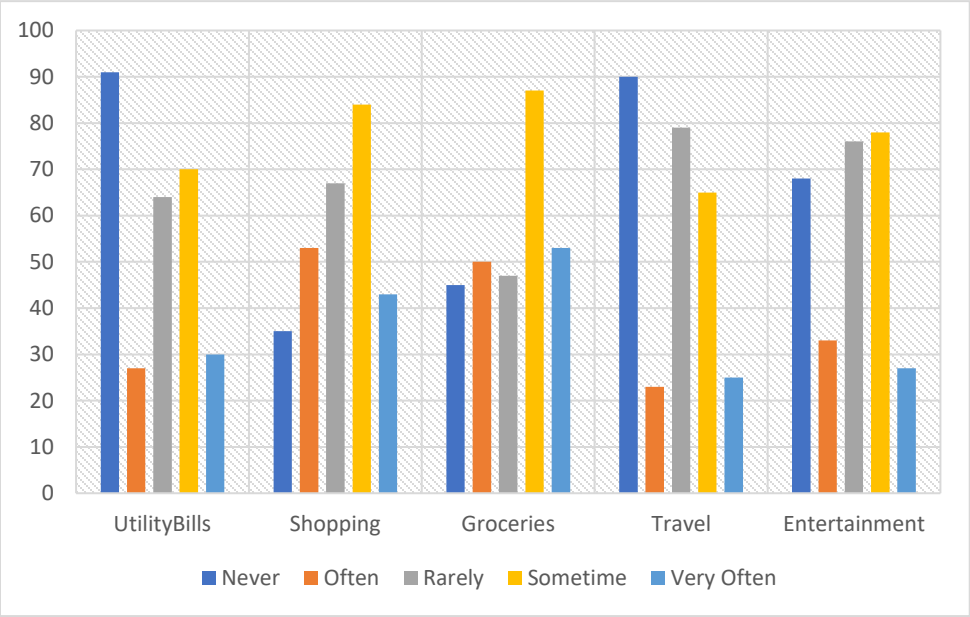



***Inferences:***

Net Banking and Paytm are the least frequently used platforms, with high "Never" responses. In contrast, Google Pay is the most preferred method, having the highest "Very Often" usage. Credit/Debit Card usage is distributed, showing consistent "Sometimes" and "Often" usage, while PhonePe and Other UPI apps have moderate usage, with most responses in "Never" and "Rarely" categories.

### *3.1.10 Frequency of Usage Across Different Activities*

| Usage | UtilityBills | Shopping | Groceries | Travel | Entertainment |
|---|---|---|---|---|---|
| **Never** | 91 | 35 | 45 | 90 | 68 |
| **Often** | 27 | 53 | 50 | 23 | 33 |
| **Rarely** | 64 | 67 | 47 | 79 | 76 |
| **Sometime** | 70 | 84 | 87 | 65 | 78 |
| **Very Often** | 30 | 43 | 53 | 25 | 27 |

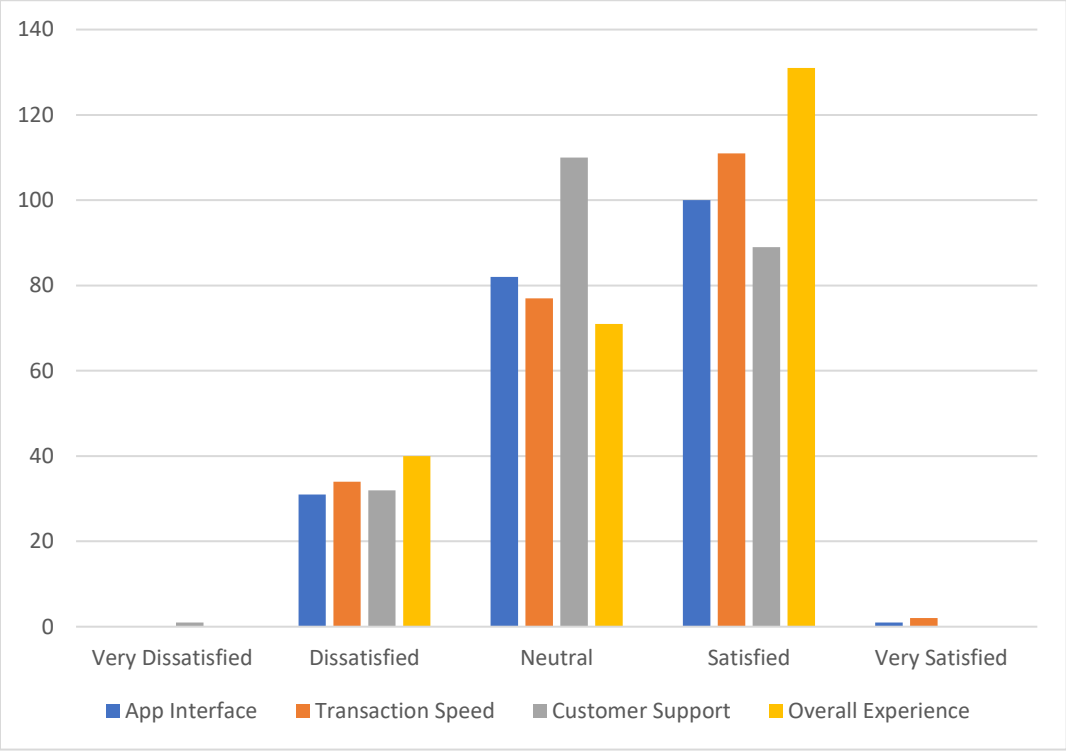



***Inferences:***

The chart indicates that digital payment usage is lowest for utility bills and travel, with a higher frequency of "Never" responses. In contrast, categories like shopping, groceries, and entertainment show a higher tendency for occasional or rare use, suggesting varying adoption patterns across different spending categories.

### *3.1.11 User Satisfaction Levels Across Digital Payment Features*

| Usage | App Interface | Transaction Speed | Customer Support | Overall Experience |
|---|---|---|---|---|
| **Very Dissatisfied** | 0 | 0 | 1 | 0 |
| **Dissatisfied** | 31 | 34 | 32 | 40 |
| **Neutral** | 82 | 77 | 110 | 71 |
| **Satisfied** | 100 | 111 | 89 | 131 |
| **Very Satisfied** | 1 | 2 | 0 | 0 |

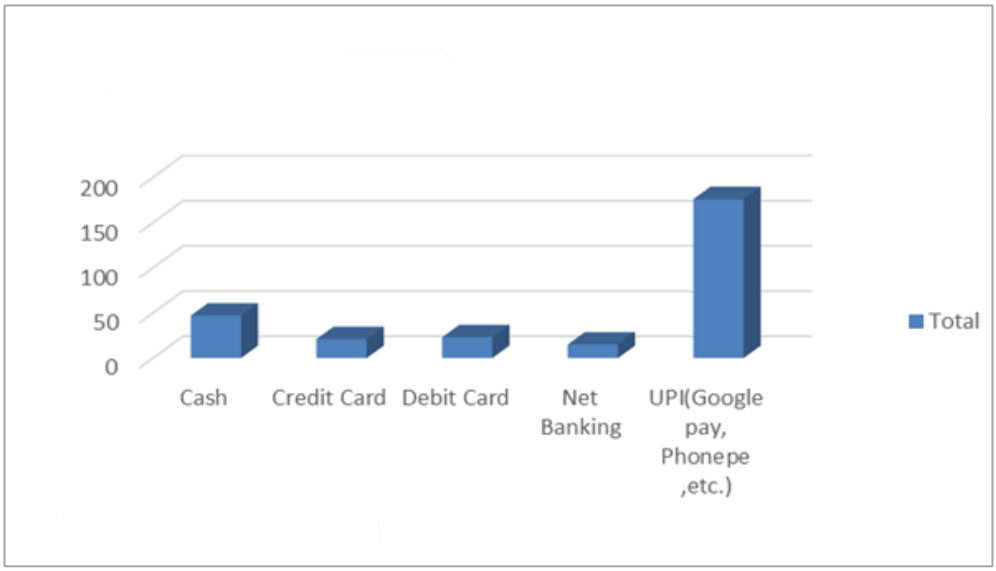



***Inferences:***

Most users are satisfied with app interface, transaction speed, and overall experience, while customer support received a higher neutral response, indicating mixed feedback. Very few users are very dissatisfied or very satisfied, showing moderate overall sentiment.

### *3.1.12 Preferred Payment Methods for High-Value Transactions (Above ₹10,000)*

| Row Labels | Count of Preferred_HighValue_Payment |
|---|---|
| Cash | 47 |
| Credit Card | 21 |
| Debit Card | 23 |
| Net Banking | 15 |
| UPI(Google pay, Phonepe ,etc.) | 176 |
| Grand Total | **282** |

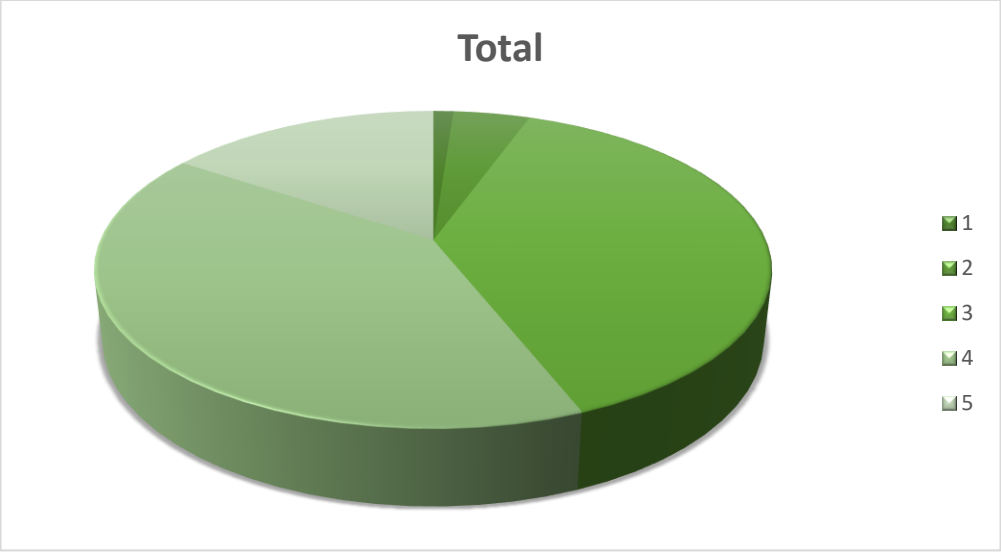



***Inferences:***

The graph shows that UPI (Google Pay, PhonePe, etc.) is the most preferred payment method for high-value transactions above ₹10,000, with 176 responses, indicating a strong preference for fast and convenient digital payments. Cash stands as the second choice, suggesting that a notable portion of people still rely on traditional methods. Credit cards, debit cards, and net banking are less preferred, showing a shift away from these options for high-value transactions.

### *3.1.13 Sum of Trust Level*

| Row Labels | Sum of Trust Level |
|---|---|
| **1** | 11 |
| **2** | 42 |
| **3** | 369 |
| **4** | 388 |
| **5** | 150 |
| **Grand Total** | **960** |

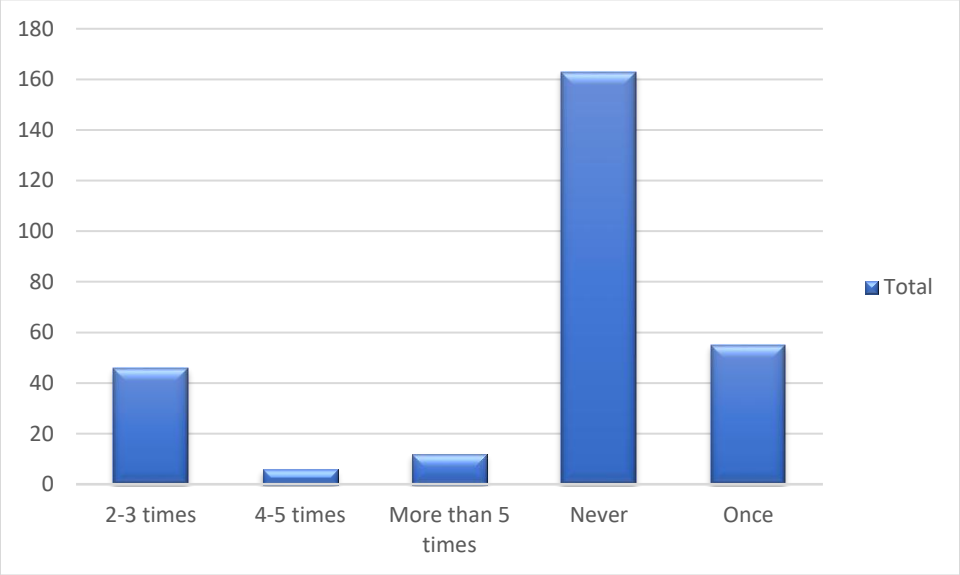



***Inferences:***

Based on the sums, most trust ratings cluster around 3 and 4, indicating a generally moderate-to-high trust level. Ratings of 1 (lowest) are very rare, and 5 (highest) is less common but still notable. Overall, this distribution suggests that while trust is largely positive, there is room to elevate it further toward the highest level.

### *3.1.14 Count of Security Incidents*

| Row Labels | Count of Security Incidents |
|---|---|
| **2-3 times** | 46 |
| **4-5 times** | 6 |
| **More than 5 times** | 12 |
| **Never** | 163 |
| **Once** | 55 |
| **Grand Total** | **282** |

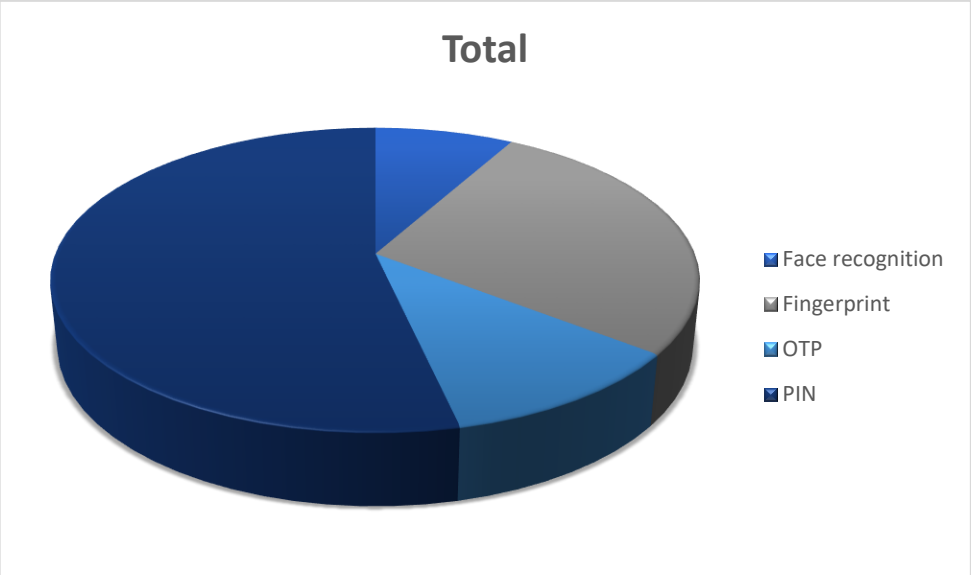



***Inferences:***

Most respondents (163 out of 282) report never experiencing a security incident, while another significant group (52) report experiencing it once. Fewer respondents (36, 13, and 6) have experienced incidents 2–3 times, 4–5 times, or more than 5 times, respectively. Overall, security incidents appear relatively rare, though a minority do encounter repeated issues.

### *3.1.15 Count of Preferred Authentication*

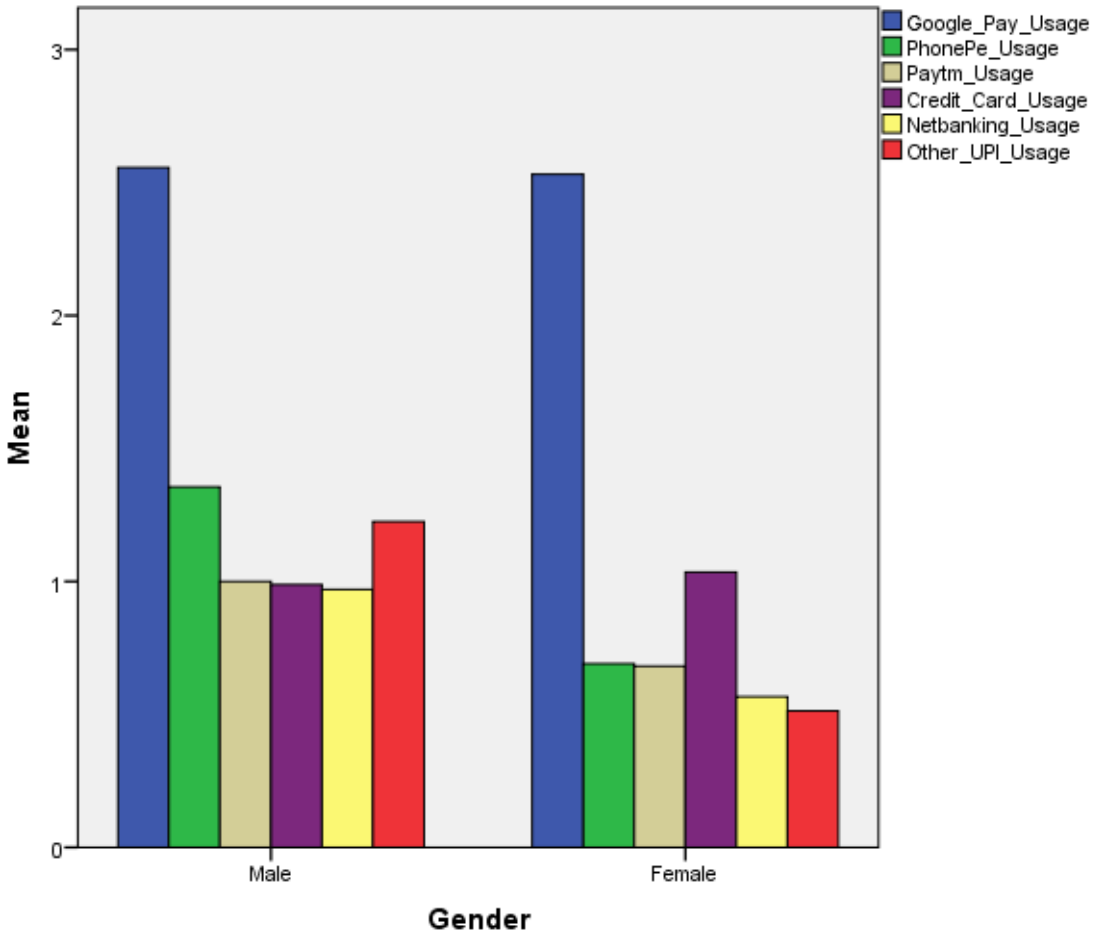
| Row Labels | Count of Preferred Authentication |
|---|---|
| **Face recognition** | 23 |
| **Fingerprint** | 77 |
| **OTP** | 31 |
| **PIN** | 151 |
| **Grand Total** | **282** |




***Inferences:***

Most respondents (151 out of 282) prefer using a PIN, making it the most popular choice. Fingerprint is the second-most common (77), followed by OTP (31). Face recognition is the least preferred (23). This suggests traditional methods (PIN) remain dominant, while biometrics and OTP see moderate adoption.

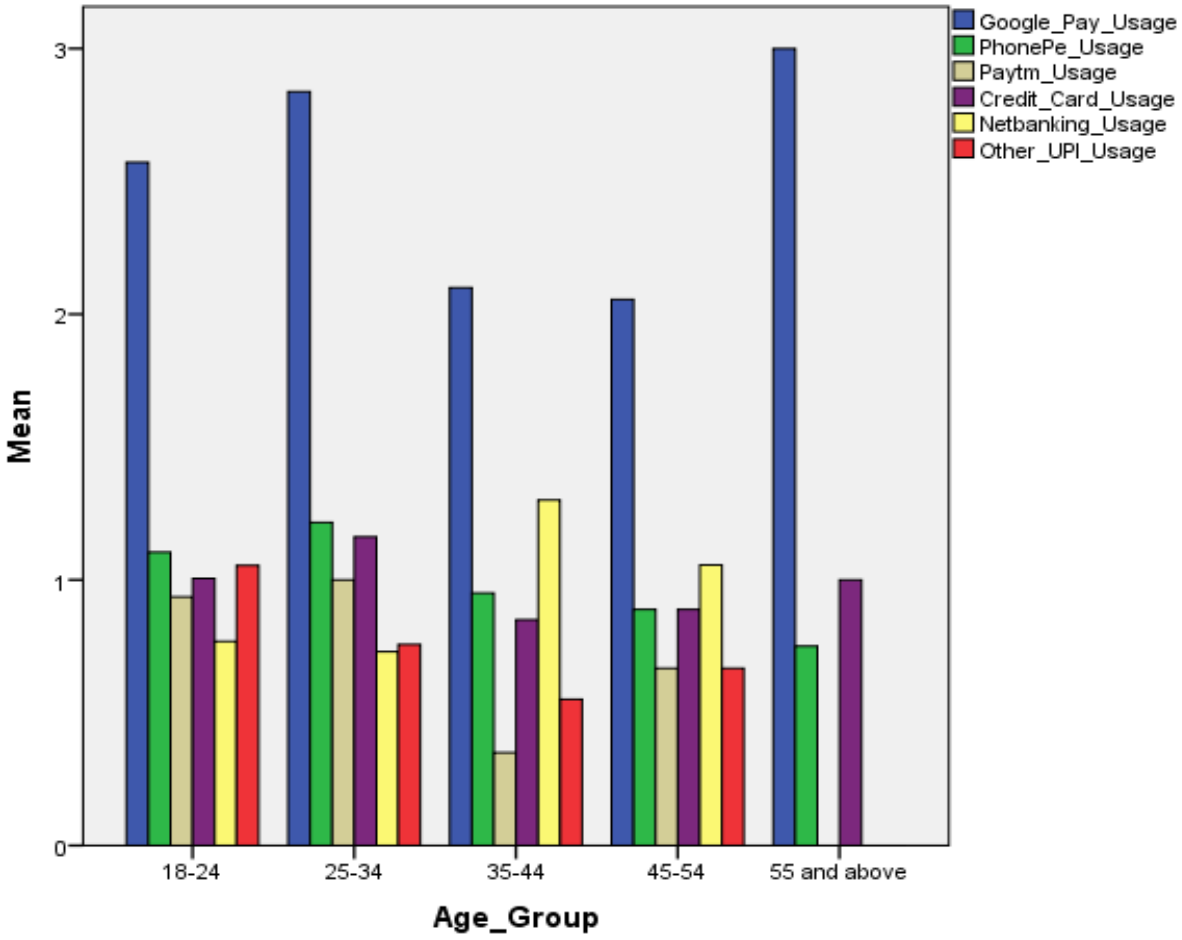
### *3.1.16 Gender-wise Platform Preference*




***Inferences:***

Google Pay is the top choice for both males and females.Males prefer a wider range of platforms, especially PhonePe and Other UPI apps.Females rely more on Google Pay and Credit Cards.Gender differences highlight varied preferences, useful for targeted promotions.

### *3.1.17 Age Group-wise Platform Preference*




***Inferences:***

Google Pay is the most used platform across all age groups, while PhonePe and Paytm are popular among youth. Credit Card and Netbanking are used more by middle-aged groups. Usage patterns vary with age.

## 3.2 Statistical Analysis

**Chi-square Test**

### *3.2.1 Analysis of Gender Influence on Experienced Security Incidents*

*Hypothesis*

$H_0$: There is no association between Gender and experiencing security incidents.

$H_1$: there is an association between Gender and experiencing security incidents.

**Gender * security_binary Crosstabulation**

| | | | security_binary | | Total |
|---|---|---|---|---|---|
| | | | .00 | 1.00 | |
| Gender | Male | Count | 88 | 81 | 169 |
| | | Expected Count | 97.7 | 71.3 | 169.0 |
| | Female | Count | 75 | 38 | 113 |
| | | Expected Count | 65.3 | 47.7 | 113.0 |
| Total | | Count | 163 | 119 | 282 |
| | | Expected Count | 163.0 | 119.0 | 282.0 |

**Chi-Square Tests**

| | Value | df | Asymp. Sig. (2-sided) |
|---|---|---|---|
| Pearson Chi-Square | 5.678[a] | 1 | .017 |
| Likelihood Ratio | 5.738 | 1 | .017 |
| N of Valid Cases | 282 | | |

***Inferences:***

The p-value is less than 0.05, so we reject the null hypothesis .There is a significant association between Gender and Experienced Security Incidents ($\chi2(1) = 5.678, p=0.017$).

### *3.2.2 Examining the Relationship Between Trust Levels and Education Levels*

*Hypothesis*

$H_0$: There is no association between education level and trust level.

$H_1$: There is an association between education level and trust level.

**Education * Trust_Level Crosstabulation**

| | | | Trust_Level | | | | | Total |
|---|---|---|---|---|---|---|---|---|
| | | | Very Low | Low | Neutral | High | Very High | |
| Education | High School | Count | 4 | 3 | 8 | 16 | 9 | 40 |
| | | Expected Count | 1.6 | 3.0 | 17.4 | 13.8 | 4.3 | 40.0 |
| | Undergraduate | Count | 4 | 9 | 55 | 38 | 5 | 111 |
| | | Expected Count | 4.3 | 8.3 | 48.4 | 38.2 | 11.8 | 111.0 |
| | Postgraduate | Count | 0 | 6 | 33 | 22 | 12 | 73 |
| | | Expected Count | 2.8 | 5.4 | 31.8 | 25.1 | 7.8 | 73.0 |
| | Professional Degree | Count | 2 | 3 | 26 | 19 | 2 | 52 |
| | | Expected Count | 2.0 | 3.9 | 22.7 | 17.9 | 5.5 | 52.0 |
| | Others | Count | 1 | 0 | 1 | 2 | 2 | 6 |
| | | Expected Count | .2 | .4 | 2.6 | 2.1 | .6 | 6.0 |
| Total | | Count | 11 | 21 | 123 | 97 | 30 | 282 |
| | | Expected Count | 11.0 | 21.0 | 123.0 | 97.0 | 30.0 | 282.0 |

**Chi-Square Tests**

| | Value | df | Asymp. Sig. (2-sided) |
|---|---|---|---|
| Pearson Chi-Square | 35.006[a] | 16 | .004 |
| Likelihood Ratio | 36.715 | 16 | .002 |
| Linear-by-Linear Association | .011 | 1 | .915 |
| N of Valid Cases | 282 | | |

a. 12 cells (48.0%) have expected count less than 5. The minimum expected count is .23.

***Inferences:***

The p-value (0.004) is less than 0.05, so we reject the null hypothesis. There is a statistically significant association between education level and trust level.

### *3.2.3 Association between Gender and Usage Frequency of Digital Payment*

*Hypothesis*

$H_0$: There is no significant association between Gender and Usage Frequency of Digital Payments.

$H_1$: There is a significant association between Gender and Usage Frequency of Digital Payments.

**Gender * Usage_Frequency Crosstabulation**

| | | | Usage_Frequency | | | | | Total |
|---|---|---|---|---|---|---|---|---|
| | | | Rarely | Occasionally | Moderately | Frequently | Very Frequently | |
| Gender | Male | Count | 12 | 13 | 23 | 42 | 79 | 169 |
| | | Expected Count | 14.4 | 15.6 | 25.8 | 53.3 | 59.9 | 169.0 |
| | Female | Count | 12 | 13 | 20 | 47 | 21 | 113 |
| | | Expected Count | 9.6 | 10.4 | 17.2 | 35.7 | 40.1 | 113.0 |
| Total | | Count | 24 | 26 | 43 | 89 | 100 | 282 |
| | | Expected Count | 24.0 | 26.0 | 43.0 | 89.0 | 100.0 | 282.0 |

**Chi-Square Tests**

| | Value | df | Asymp. Sig. (2-sided) |
|---|---|---|---|
| Pearson Chi-Square | 23.954[a] | 4 | .000 |
| Likelihood Ratio | 25.134 | 4 | .000 |
| Linear-by-Linear Association | 10.826 | 1 | .001 |
| N of Valid Cases | 282 | | |

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 9.62.

***Inferences:***

Since the p-value = 0.000 < 0.05, we reject $H_0$. Therefore, there is a significant association between Gender and Usage Frequency of Digital Payments. This indicates that the frequency with which people use digital payments depends on their gender.

## Correlation

### *3.2.4 Effect of Digital Payments on Personal Financial Management*

*Tests of Normality*

Before conducting the correlation analysis, a normality test was performed using the Kolmogorov-Smirnov and Shapiro-Wilk tests.

**Tests of Normality**

| | Kolmogorov-Smirnov[a] | | | Shapiro-Wilk | | |
|---|---|---|---|---|---|---|
| | Statistic | df | Sig. | Statistic | df | Sig. |
| Usage_Frequency | .245 | 282 | .000 | .834 | 282 | .000 |
| Monthly_Transactions | .368 | 282 | .000 | .687 | 282 | .000 |
| Expense_Tracking_Benefit | .215 | 282 | .000 | .881 | 282 | .000 |
| Savings_Increase | .197 | 282 | .000 | .889 | 282 | .000 |
| Impulse_Purchases | .224 | 282 | .000 | .878 | 282 | .000 |
| Bill_Payment_Benefit | .295 | 282 | .000 | .849 | 282 | .000 |

a. Lilliefors Significance Correction

The results indicate that the data does not follow a normal distribution ($p < 0.05$ for multiple variables). Therefore, Spearman's correlation (a non-parametric alternative to Pearson's correlation) was used.

*Hypothesis*

$H_0$: There is no significant correlation between digital payment usage and financial habits.

$H_1$: There is a significant correlation between digital payment usage and financial habits.

**Correlations**

| | | | Usage_Frequency | Monthly_Transactions | Expense_Tracking_Benefit | Savings_Increase | Impulse_Purchases | Bill_Payment_Benefit |
|---|---|---|---|---|---|---|---|---|
| Spearman's rho | Usage_Frequency | Correlation Coefficient | 1.000 | .364** | .015 | -.084 | .069 | .089 |
| | | Sig. (2-tailed) | . | .000 | .803 | .161 | .248 | .137 |
| | | N | 282 | 282 | 282 | 282 | 282 | 282 |
| | Monthly_Transactions | Correlation Coefficient | .364** | 1.000 | .008 | -.004 | .037 | .095 |
| | | Sig. (2-tailed) | .000 | . | .898 | .943 | .541 | .113 |
| | | N | 282 | 282 | 282 | 282 | 282 | 282 |
| | Expense_Tracking_Benefit | Correlation Coefficient | .015 | .008 | 1.000 | .201** | .306** | .326** |
| | | Sig. (2-tailed) | .803 | .898 | . | .001 | .000 | .000 |
| | | N | 282 | 282 | 282 | 282 | 282 | 282 |
| | Savings_Increase | Correlation Coefficient | -.084 | -.004 | .201** | 1.000 | .039 | .045 |
| | | Sig. (2-tailed) | .161 | .943 | .001 | . | .520 | .450 |
| | | N | 282 | 282 | 282 | 282 | 282 | 282 |
| | Impulse_Purchases | Correlation Coefficient | .069 | .037 | .306** | .039 | 1.000 | .324** |
| | | Sig. (2-tailed) | .248 | .541 | .000 | .520 | . | .000 |
| | | N | 282 | 282 | 282 | 282 | 282 | 282 |
| | Bill_Payment_Benefit | Correlation Coefficient | .089 | .095 | .326** | .045 | .324** | 1.000 |
| | | Sig. (2-tailed) | .137 | .113 | .000 | .450 | .000 | . |
| | | N | 282 | 282 | 282 | 282 | 282 | 282 |

**. Correlation is significant at the 0.01 level (2-tailed).

***Inferences:***

There is a significant positive correlation between Usage Frequency and Monthly Transactions (r = 0.364, p = 0.000), indicating that higher usage frequency is associated with more monthly transactions.

Expense Tracking Benefit is positively correlated with Savings Increase (r = 0.201, p = 0.001), Impulse Purchases (r = 0.306, p = 0.000), and Bill Payment Benefit (r = 0.326, p = 0.000), suggesting that users who track expenses tend to save more, make more impulse purchases, and manage bill payments better.

Additionally, Impulse Purchases are positively correlated with Bill Payment Benefit (r = 0.324, p = 0.000), indicating that those who indulge in impulse buying also report better bill payment experiences.

However, Usage Frequency shows no significant correlation with Expense Tracking Benefit, Savings Increase, or Bill Payment Benefit. Also, Savings Increase and Impulse Purchases (r = 0.039, p = 0.520) are not significantly related, indicating that higher savings do not necessarily lead to reduced impulse buying.

## Mann-Whitney U Test

### *3.2.5 Assessing Gender Differences in Usage Frequency*

*Tests of Normality*

The Kolmogorov-Smirnov and Shapiro-Wilk tests were used to assess normality for Usage_Frequency across genders (Table 1).

*Table 1: Tests of Normality*

**Tests of Normality**

| | | Kolmogorov-Smirnov[a] | | | Shapiro-Wilk | | |
|---|---|---|---|---|---|---|---|
| | Gender | Statistic | df | Sig. | Statistic | df | Sig. |
| Usage_Frequency | Male | .264 | 169 | .000 | .786 | 169 | .000 |
| | Female | .272 | 113 | .000 | .864 | 113 | .000 |

a. Lilliefors Significance Correction

The p-values (Sig.) = .000 for both tests, indicating non-normal distribution for both genders ($p < 0.05$). Since normality is violated, a non-parametric test (Mann-Whitney U) will be used instead of a t-test.

*Hypothesis*

$H_0$: There is no significant difference in Usage Frequency between Males and Females.

$H_1$: There is a significant difference in Usage Frequency between Males and Females.

**Ranks**

| | Gender | N | Mean Rank | Sum of Ranks |
|---|---|---|---|---|
| Usage_Frequency | Male | 169 | 156.55 | 26457.00 |
| | Female | 113 | 118.99 | 13446.00 |
| | Total | 282 | | |

**Test Statistics[a]**

| | Usage_Frequency |
|---|---|
| Mann-Whitney U | 7005.000 |
| Wilcoxon W | 13446.000 |
| Z | -3.953 |
| Asymp. Sig. (2-tailed) | .000 |

a. Grouping Variable: Gender

***Inferences:***

Since p-value (.000) < 0.05, we reject $H_0$, indicating a significant difference in Usage Frequency between Males and Females. The higher mean rank for Males (156.55) suggests they have a higher Usage Frequency than Females (118.99).

### *3.2.6 Gender Differences in Overall Satisfaction*

*Tests of Normality*

The Kolmogorov-Smirnov and Shapiro-Wilk tests were used to assess the normality of Overall Satisfaction scores across genders.

**Tests of Normality**

| | | Kolmogorov-Smirnov[a] | | | Shapiro-Wilk | | |
|---|---|---|---|---|---|---|---|
| | Gender | Statistic | df | Sig. | Statistic | df | Sig. |
| Satisfaction_Overall | Male | .267 | 169 | .000 | .855 | 169 | .000 |
| | Female | .294 | 113 | .000 | .847 | 113 | .000 |

a. Lilliefors Significance Correction

The p-values (Sig.) being less than 0.05 indicate non-normal distribution of Overall Satisfaction across genders. Hence, the Mann-Whitney U test, a non-parametric alternative to the t-test, was applied.

*Hypothesis*

$H_0$: There is no significant difference in Overall Satisfaction between Males and Females.

$H_1$: There is a significant difference in Overall Satisfaction between Males and Females.

**Ranks**

| | Gender | N | Mean Rank | Sum of Ranks |
|---|---|---|---|---|
| Satisfaction_Overall | Male | 169 | 135.22 | 22853.00 |
| | Female | 113 | 150.88 | 17050.00 |
| | Total | 282 | | |

**Test Statistics[a]**

| | Satisfaction_ Overall |
|---|---|
| Mann-Whitney U | 8488.000 |
| Wilcoxon W | 22853.000 |
| Z | -1.684 |
| Asymp. Sig. (2-tailed) | .092 |

a. Grouping Variable: Gender

***Inferences:***

Since the p-value (0.092) is greater than 0.05, we fail to reject the null hypothesis ($H_0$). Hence, there is no significant difference in overall satisfaction between males and females.

### *3.2.7 Trust Levels Among Private and Government Employees*

*Tests of Normality*

The Kolmogorov-Smirnov and Shapiro-Wilk tests were used to assess normality for Trust_Level across occupations.

**Tests of Normality**

| | | Kolmogorov-Smirnov[a] | | | Shapiro-Wilk | | |
|---|---|---|---|---|---|---|---|
| | Occupation | Statistic | df | Sig. | Statistic | df | Sig. |
| Trust_Level | Private Sector Employee | .296 | 44 | .000 | .835 | 44 | .000 |
| | Government Employee | .272 | 15 | .004 | .870 | 15 | .034 |

a. Lilliefors Significance Correction

The Private Sector Employees exhibit a non-normal distribution (p = .000 for both tests). For Government Employees, Kolmogorov-Smirnov test indicates non-normality (p = .004), while Shapiro-Wilk suggests marginal normality (p = .034). Due to at least one test showing non-normality, the Mann-Whitney U test is used instead of a t-test.

*Hypothesis*

$H_0$: There is no significant difference in Trust Level between Private Sector and Government Employees.

$H_1$: There is a significant difference in Trust Level between Private Sector and Government Employees.

**Ranks**

| | Occupation | N | Mean Rank | Sum of Ranks |
|---|---|---|---|---|
| Trust_Level | Private Sector Employee | 44 | 29.35 | 1291.50 |
| | Government Employee | 15 | 31.90 | 478.50 |
| | Total | 59 | | |

**Test Statistics**[a]

| | Trust_Level |
|---|---|
| Mann-Whitney U | 301.500 |
| Wilcoxon W | 1291.500 |
| Z | -.546 |
| Asymp. Sig. (2-tailed) | .585 |

a. Grouping Variable: Occupation

***Inferences:***

Since the p-value (.585) > 0.05, we fail to reject $H_0$, indicating no significant difference in Trust Level between Private Sector and Government Employees.

## Kruskal-Wallis Test

### *3.2.8 Effect of Usage Duration on Overall Satisfaction*

*Tests of Normality*

Before conducting the Kruskal-Wallis test, a normality test was performed using the Kolmogorov-Smirnov and Shapiro-Wilk tests.

**Tests of Normality**

| | | Kolmogorov-Smirnov[a] | | | Shapiro-Wilk | | |
|---|---|---|---|---|---|---|---|
| | Usage_Duration | Statistic | df | Sig. | Statistic | df | Sig. |
| Satisfaction_Overall | Less than 6 months | .191 | 71 | .000 | .893 | 71 | .000 |
| | 6 months - 1 year | .238 | 37 | .000 | .866 | 37 | .000 |
| | 1-2 years | .261 | 61 | .000 | .883 | 61 | .000 |
| | 2-4 years | .387 | 67 | .000 | .747 | 67 | .000 |
| | More than 4 years | .292 | 46 | .000 | .763 | 46 | .000 |

a. Lilliefors Significance Correction

Since the p-values are all < 0.05, the assumption of normality is violated. Therefore, the Kruskal-Wallis test, a non-parametric alternative to ANOVA, was used.

*Hypothesis*

$H_0$:There is no significant difference in Satisfaction_Overall across different Usage_Duration groups.

$H_1$: There is a significant difference in Satisfaction_Overall across different Usage_Duration groups.

**Ranks**

| | Usage_Duration | N | Mean Rank |
|---|---|---|---|
| Satisfaction_Overall | Less than 6 months | 71 | 112.15 |
| | 6 months - 1 year | 37 | 129.57 |
| | 1-2 years | 61 | 158.35 |
| | 2-4 years | 67 | 169.64 |
| | More than 4 years | 46 | 133.05 |
| | Total | 282 | |

**Test Statistics[a,b]**

| | Satisfaction_ Overall |
|---|---|
| Chi-Square | 23.930 |
| df | 4 |
| Asymp. Sig. | .000 |

a. Kruskal Wallis Test

b. Grouping Variable: Usage_Duration

***Inferences:***

Since the p-value (.000) < 0.05, we reject the null hypothesis, indicating that there is a significant difference in Satisfaction_Overall across different Usage_Duration groups.

### *3.2.9 Analyzing the Relationship Between Monthly Income and Transaction Frequency*

*Tests of Normality*

The Kolmogorov-Smirnov and Shapiro-Wilk tests were used to assess normality for Average Monthly Transactions across the levels of Monthly Income.

**Tests of Normality**

| | Kolmogorov-Smirnov[a] | | | Shapiro-Wilk | | |
|---|---|---|---|---|---|---|
| | Statistic | df | Sig. | Statistic | df | Sig. |
| Monthly_Income | .288 | 282 | .000 | .754 | 282 | .000 |
| Monthly_Transactions | .368 | 282 | .000 | .687 | 282 | .000 |

a. Lilliefors Significance Correction

The p-values (Sig.) = .000 for both tests, indicating non-normal distribution for Average Monthly Transactions across the levels of Monthly Income ($p < 0.05$).Since normality is violated, a non-parametric test (Kruskal-Wallis test)will be used instead of a ANOVA.

*Hypothesis*

$H_0$: The distribution of Average Monthly Transactions is the same across all levels of Monthly Income.

$H_1$: The distribution of Average Monthly Transactions is not the same across all levels of Monthly Income.

**Ranks**

| | Monthly_Income | N | Mean Rank |
|---|---|---|---|
| Monthly_Transactions | Below 5,000 | 101 | 66.29 |
| | 5000-15000 | 29 | 82.28 |
| | 15001-30000 | 8 | 79.38 |
| | Above 30000 | 4 | 109.25 |
| | Total | 142 | |

**Test Statistics[a,b]**

| | Monthly_Transactions |
|---|---|
| Chi-Square | 8.261 |
| df | 3 |
| Asymp. Sig. | .041 |

a. Kruskal Wallis Test

b. Grouping Variable: Monthly_Income

***Inferences:***

Since the p-value ($p = 0.041$) from the Kruskal-Wallis test ($\chi^2 = 8.261$, $df = 3$) is less than 0.05, the null hypothesis is rejected.There is a significant difference in Monthly Transactions across different Monthly Income levels.

### *3.2.10 Impact of Transaction Speed on Overall Satisfaction*

*Tests of Normality*

The Kolmogorov-Smirnov and Shapiro-Wilk tests were used to assess normality for Overall Satisfaction across the levels of Transaction Speed Satisfaction.

**Tests of Normality**

| | Kolmogorov-Smirnov[a] | | | Shapiro-Wilk | | |
|---|---|---|---|---|---|---|
| | Statistic | df | Sig. | Statistic | df | Sig. |
| Satisfaction_Transaction_Speed | .239 | 282 | .000 | .886 | 282 | .000 |
| Satisfaction_Overall | .279 | 282 | .000 | .852 | 282 | .000 |

a. Lilliefors Significance Correction

The p-values (Sig.) = .000 for both tests, indicating non-normal distribution for Overall Satisfaction across the levels of Transaction Speed Satisfaction. ($p < 0.05$). Since normality is violated, a non-parametric test (Kruskal-Wallis test)will be used instead of a ANOVA.

*Hypothesis*

$H_0$: There is no significant difference in Overall Satisfaction across the levels of Transaction Speed Satisfaction.

$H_1$: There is a significant difference in Overall Satisfaction across the levels of Transaction Speed Satisfaction.

**Ranks**

| | Satisfaction_Transaction_Speed | N | Mean Rank |
|---|---|---|---|
| Satisfaction_Overall | Very Dissatisfied | 26 | 46.81 |
| | Dissatisfied | 34 | 49.29 |
| | Neutral | 77 | 120.88 |
| | Satisfied | 111 | 186.02 |
| | Very Satisfied | 34 | 207.47 |
| | Total | 282 | |

**Test Statistics**[a,b]

| | Satisfaction_Overall |
|---|---|
| Chi-Square | 157.665 |
| df | 4 |
| Asymp. Sig. | .000 |

a. Kruskal Wallis Test

b. Grouping Variable: Satisfaction_Transaction_Speed

***Inferences:***

Since the p-value from the Kruskal-Wallis test ($\chi^2$ = 157.665, df = 4) is less than 0.05, the null hypothesis is rejected. There is a significant difference in Overall Satisfaction across different levels of Transaction Speed Satisfaction.

# CHAPTER 4

# CONCLUSION

This study on "Digital Payment Usage Patterns and Financial Behaviour: A Statistical Analysis of Consumer Demographics" highlights that demographic factors such as age, gender, education, occupation, and income significantly influence the usage, trust, and adoption of digital payments. The analysis shows that young individuals, particularly students, are the primary users, with Google Pay as the most preferred platform and PIN as the most trusted authentication method.

Statistical tests confirmed significant associations between gender and security incidents, education and trust, and gender and usage frequency, indicating that user experiences vary across groups. Frequent users are more likely to track expenses and pay bills on time, but also tend to make impulse purchases, reflecting mixed financial behaviour.

Although users are satisfied with app interface and transaction speed, trust and customer support remain concerns. Moreover, usage duration, income, and transaction speed significantly influence overall satisfaction.

In conclusion, while digital payments are widely embraced for their convenience and speed, efforts to enhance security, trust, and customer service are essential to ensure broader and safer adoption. These insights are useful for digital payment providers, financial institutions, and policymakers to create a secure and inclusive digital payment ecosystem.

# BIBLIOGRAPHY


1. Gupta, S. C., & Kapoor, V. K. (2020). *Fundamentals of mathematical statistics* (12th ed.). Sultan Chand & Sons.
2. Kothari, C. R. (2004). *Research methodology: Methods and techniques* (2nd ed.). New Age International Publishers.
3. National Payments Corporation of India. (2023). *UPI product statistics and reports*. https://www.npci.org.in
4. Pallant, J. (2020). *SPSS survival manual: A step-by-step guide to data analysis using IBM SPSS* (7th ed.). McGraw-Hill Education.
5. Reserve Bank of India. (2023). *Report on trends and progress of banking in India*. https://www.rbi.org.in
6. Laerd Statistics. (n.d.). *SPSS statistical tutorials and guides*. https://statistics.laerd.com/

## QUESTIONNAIRE

# Digital Payment Usage Patterns and Financial Behavior

We are final year BSc.Statistics students from CHMKM Government Arts and Science College, Koduvally, conducting a study on digital payment patterns and financial behavior as part of our academic project. Led by Sreya, Akhil ,Sourav and Thahira husna, this 5-minute survey aims to understand digital payment usage in everyday life. Your responses will be kept confidential and used solely for academic purposes. For queries: husnahus9526@gmail.com Thank you for your participation.

* Indicates required question

### Section A: Demographic Information

1.Age Group: * *Mark only one oval.*

- 18-25
- 25-34
- 35-44
- 45-54
- 55 and above

2.Gender: * *Mark only one oval.*

- Male
- Female
- Other
- Prefer not to say

3. Educational Qualification: * *Mark only one oval.*

- High School
- Undergraduate
- Postgraduate
- Professional Degree
- Other:

4. Occupation: * *Mark only one oval.*

- Student
- Private Sector Employee
- Government Employee
- Self-employed/Business
- Homemaker
- Retired
- Other: _____

5. Monthly Income (in ₹): * *Mark only one oval.*

- Below 25,000
- 25,000-50,000
- 50,001-75,000
- Above 75,000
- Not Applicable

**Section B: Digital Payment Usage and Transaction Types**

6. How long have you been using digital payment methods? * *Mark only one oval.*

- Less than 6 months
- 6 months - 1 year
- 1-2 years
- 2-4 years
- More than 4 years

7. Frequency of digital payment usage: * *Mark only one oval.*

- Very Frequently (Multiple times per day)
- Frequently (Daily or almost daily)
- Moderately (2-3 times per week)
- Occasionally (1-2 times per month)
- Rarely (Less than once a month)

8. Average monthly digital transactions (in ₹): * *Mark only one oval.*

- Below 5,000
- 5,000-15,000
- 15,001-30,000
- Above 30,000

9. Rate your usage frequency for these platforms: * *Mark only one oval per row.*

| | Never | Rarely | Sometimes | Often | Very often |
|---|---|---|---|---|---|
| **Google Pay** | | | | | |
| **PhonePe** | | | | | |
| **Paytm** | | | | | |
| **Other UPI apps** | | | | | |
| **Credit/Debit Cards** | | | | | |
| **Netbanking** | | | | | |

10. Rate your usage frequency for: * *Mark only one oval per row.*

| | Never | Rarely | Sometimes | Often | Very often |
|---|---|---|---|---|---|
| **Utility bills** | | | | | |
| **Shopping** | | | | | |
| **Food/Groceries** | | | | | |
| **Travel booking** | | | | | |
| **Entertainment** | | | | | |

11. What is your most preferred payment method for high-value transactions * (above ₹10,000)? *Mark only one oval.*

- UPI(Google pay, Phonepe ,etc.)
- Credit Card
- Debit Card
- Net Banking
- Cash

**Section C: Financial Management Impact**

12. Digital payments help me track my expenses better * *Mark only one oval.*

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

13. My savings have increased since using digital payments * *Mark only one oval.*

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

14. I make more impulse purchases with digital payments * *Mark only one oval.*

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

15.Digital payments help me pay bills on time * *Mark only one oval.*

- ◯ Strongly Disagree
- ◯ Disagree
- ◯ Neutral
- ◯ Agree
- ◯ Strongly Agree

**Section D: User Experience, Security and Trust**

16. Rate your satisfaction with: * *Mark only one oval per row.*

| | Very dissatisfied | Dissatisfied | Neutral | Satisfied | Very satisfied |
|---|---|---|---|---|---|
| **App interface** | ◯ | ◯ | ◯ | ◯ | ◯ |
| **Transaction speed** | ◯ | ◯ | ◯ | ◯ | ◯ |
| **Customer support** | ◯ | ◯ | ◯ | ◯ | ◯ |
| **Overall experience** | ◯ | ◯ | ◯ | ◯ | ◯ |

17. How often do you face transaction failures? * *Mark only one oval.*

- ◯ Never
- ◯ Rarely(1-2 times/month)
- ◯ Sometimes(3-5 times/month)
- ◯ Often(>5 times/month)
- ◯ Always

18. Rate your trust level in digital payments: * *(1=Very Low, 2=Low, 3=Moderate, 4=High, 5=Very High)*

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| ☆ | ☆ | ☆ | ☆ | ☆ |

19. Have you experienced any security incidents? * *Mark only one oval.*

- Never
- Once
- 2-3 times
- 4-5 times
- More than 5 times

20. Preferred authentication method: * *Mark only one oval.*

- PIN
- Fingerprint
- Face recognition
- OTP

21.In your opinion, what are the main advantages of using digital payments?(Select all that apply)

*Check all that apply.*

- [ ] Convenience and ease of use
- [ ] Expense tracking capabilities
- [ ] Rewards and cashback offers
- [ ] 24/7 accessibility
- [ ] Contactless payment option
- [ ] Speed of transactions
- [ ] Security and reduced cash handling Other:

_____

***Thank you for completing the survey!***